

Note: This document only contains the graded problem viz Exercise 5. To view the R Code used in solutions, [click here](#).

Exercise 5

In many instances, obtaining the MLE in closed-form by solving the likelihood equations $l'(\theta) = 0$ is not possible. In such cases, we need to use some numerical optimization technique to get a solution. One such technique, known as Fisher's scoring, was discussed in the class. This is an iterative method, where we successively obtain

$$\theta_{k+1} = \theta_k - \frac{l'(\theta_k)}{\mathbb{E}_\theta[l''(\theta|X_n)]|_{\theta=\theta_k}} = \theta_k + \frac{l'(\theta_k)}{I_n(\theta_k)}, \quad k = 1, 2, \dots \text{ (until convergence),}$$

where $I_n(\theta) = -\mathbb{E}_\theta[l''(\theta|X_n)]$ is the Fisher information contained in the sample $X_n := (X_1, \dots, X_n)$. Here, the derivatives are w.r.t. θ and the expectations are w.r.t. the sample X_n . In this exercise, we will derive the Fisher scoring iteration steps for one particular distribution.

Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Logistic}(\theta)$ with pdf

$$f_\theta(x) = \frac{e^{-(x-\theta)}}{(1 + e^{-(x-\theta)})^2}, \quad x \in \mathbb{R}.$$

Here $\theta \in \mathbb{R}$ is our parameter of interest.

- Write down the likelihood of θ and the likelihood equation $l'(\theta) = 0$. Does it admit a closed form solution?
- Make plots of $l(\theta)$ and $l'(\theta)$. What can you say about the existence of the MLE from these plots?
- Find $l''(\theta)$ and $I_n(\theta)$.
- Formulate the Fisher's scoring equation for finding the MLE.
- Write an iterative algorithm to find the MLE using your answer in part (d). You need to fix a threshold to determine convergence. Use a small value, e.g., 10^{-5} .
- Fix a θ and generate $n = 100$ samples from f_θ . Use your algorithm to determine the MLE for the given data.

- (g) Repeat part (e) 1000 times by generating independent random samples from the same distribution. This will give you 1000 estimated values for the parameter θ . Make a histogram of these estimates. What is your assessment of the MLE as an estimator of θ ?
- (h) Repeat parts (f) and (g) with $n = 50, 200, 500$. What do you observe?

Solution:

- (a) Given X_i 's are i.i.d, we have the following likelihood for θ

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n f_{\theta}(x_i) \\ &= \frac{\exp\{-\sum x_i + n\theta\}}{\prod (1 + e^{-(x_i - \theta)})^2} \end{aligned}$$

To make our lives easier, we can consider the log-likelihood function (essentially because log is a monotonically increasing function for positive values). Thus, we have the following

$$l(\theta) = \log(L(\theta)) = \left(-\sum_{i=1}^n x_i + n\theta\right) - 2 \sum_{i=1}^n \log(1 + e^{-(x_i - \theta)})$$

This implies,

$$\begin{aligned} l'(\theta) &= n - 2 \sum_{i=1}^n \left(\frac{1}{1 + e^{-(x_i - \theta)}} e^{-(x_i - \theta)} \right) \\ &= n - 2 \sum_{i=1}^n \left(\frac{1}{1 + e^{(x_i - \theta)}} \right) \end{aligned}$$

Thus $l'(\theta) = 0$, implies

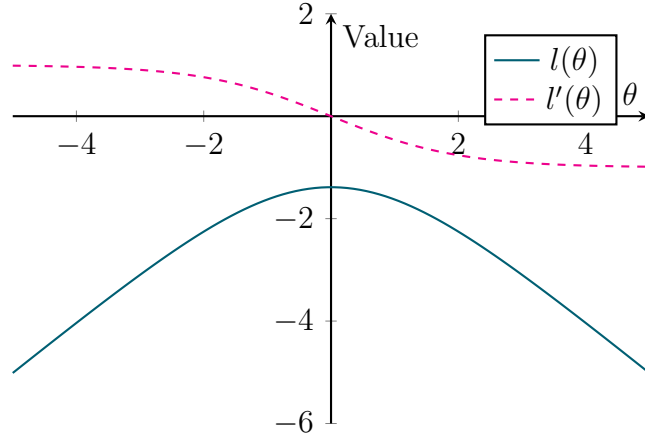
$$\sum_{i=1}^n \frac{1}{1 + e^{(x_i - \theta)}} = \frac{n}{2}$$

This equation cannot be reduced further. Thus, $l'(\theta) = 0$ clearly doesn't admit a closed form solution.

- (b) To plot $l(\theta)$ and $l'(\theta)$, we'll consider a dummy function such that $n = 1$ and $x_1 = 0$. Thus,

$$\begin{aligned} l(\theta) &= \theta - 2 \ln(1 + e^{\theta}) \\ l'(\theta) &= 1 - \frac{2e^{\theta}}{1 + e^{\theta}} = \frac{1 - e^{\theta}}{1 + e^{\theta}} \end{aligned}$$

Hence, we have the following plots



Existence of the MLE: From the plot, we can observe two critical properties:

- (a) The log-likelihood function $l(\theta)$ is strictly concave (it looks like an inverted, smooth "V" shape).
- (b) The score function $l'(\theta)$ is strictly monotonically decreasing and crosses the horizontal axis ($l'(\theta) = 0$) at exactly one point.

Because $l(\theta)$ is strictly concave everywhere, any local maximum is guaranteed to be a unique global maximum. Therefore, we can definitively say that the MLE exists and is completely unique for any given dataset.

- (c) We know that

$$l'(\theta) = n - 2 \sum_{i=1}^n \left(\frac{1}{1 + e^{(x_i - \theta)}} \right)$$

Thus, we have the following

$$l''(\theta) = -2 \sum_{i=1}^n \frac{e^{(x_i - \theta)}}{(1 + e^{(x_i - \theta)})^2}$$

Now, we'll calculate the *Fisher Information*, $I_n(\theta)$. We know that $I_n(\theta) = -\mathbb{E}(l''(\theta|X_n))$. Because $X_1, \dots, X_n$ are i.i.d., the expectation of the sum is n times the expectation of a single term. Thus, we have the following

$$\begin{aligned} I_n(\theta) &= -\mathbb{E} \left[-2 \sum_{i=1}^n \frac{e^{(X_i - \theta)}}{(1 + e^{(X_i - \theta)})^2} \right] \\ &= 2n \mathbb{E} \left[\frac{e^{(X - \theta)}}{(1 + e^{(X - \theta)})^2} \right] \\ &= 2n \int_{-\infty}^{\infty} \frac{e^{(x - \theta)}}{(1 + e^{(x - \theta)})^2} \left[\frac{e^{-(x - \theta)}}{(1 + e^{-(x - \theta)})^2} \right] dx \\ &= 2n \int_{-\infty}^{\infty} \frac{e^{2(x - \theta)}}{(1 + e^{(x - \theta)})^4} dx \end{aligned}$$

To evaluate this integral, let $u = e^{x-\theta}$. This gives $du = e^{x-\theta}dx$, and the limits of integration change from $(-\infty, \infty)$ to $(0, \infty)$. We can rewrite the numerator as $e^{2(x-\theta)}dx = e^{x-\theta} \cdot e^{x-\theta}dx = u du$. Thus we have the following,

$$I_n(\theta) = 2n \int_0^\infty \frac{u}{(1+u)^4} du$$

Let $v = 1 + u$, which implies $u = v - 1$ and $du = dv$. The limits now change to $(1, \infty)$,

$$\begin{aligned} I_n(\theta) &= 2n \int_1^\infty \frac{v-1}{v^4} dv \\ &= 2n \int_1^\infty (v^{-3} - v^{-4}) dv \\ &= 2n \left[\frac{v^{-2}}{-2} - \frac{v^{-3}}{-3} \right]_1^\infty \\ &= 2n \left((0 - 0) - \left(\frac{1}{-2} - \frac{1}{-3} \right) \right) \\ &= 2n \left(\frac{1}{2} - \frac{1}{3} \right) \\ &= 2n \left(\frac{1}{6} \right) = \frac{n}{3} \end{aligned}$$

Thus, the Fisher Information for the sample is,

$$\boxed{I_n(\theta) = \frac{n}{3}}$$

(d) We have the following *Fisher's Scoring* formula

$$\theta_{k+1} = \theta_k + \frac{l'(\theta_k)}{I_n(\theta_k)}$$

Thus, for the given logistic distribution, we have the following

$$\begin{aligned} \theta_{k+1} &= \theta_k + \frac{3}{n} \left(n - 2 \sum_{i=1}^n \left(\frac{1}{1 + e^{(x_i - \theta_k)}} \right) \right) \\ &= \theta_k + 3 - \frac{6}{n} \sum_{i=1}^n \left(\frac{1}{1 + e^{(x_i - \theta_k)}} \right) \end{aligned}$$

(e) Consider the below Pseudocode and its R code implementation

Pseudocode:

1. **Input:** A sample dataset $x = (x_1, x_2, \dots, x_n)$, tolerance $\epsilon = 10^{-5}$, and maximum iterations $N_{\max}$.

2. **Initialize:** Set the initial guess $\theta_0 = \frac{1}{n} \sum_{i=1}^n x_i$ (sample mean). Set $k = 0$ and error $= \infty$.
3. **While** (error $> \epsilon$) and ($k < N_{\max}$) **do:**
 - i. Compute the sum term: $S_k = \sum_{i=1}^n \left(\frac{1}{1 + e^{(x_i - \theta_k)}} \right)$
 - ii. Update the estimate: $\theta_{k+1} = \theta_k + 3 - \frac{6}{n} S_k$
 - iii. Calculate the error: error $= |\theta_{k+1} - \theta_k|$
 - iv. Update for next iteration: $\theta_k \leftarrow \theta_{k+1}$ and $k \leftarrow k + 1$
4. **Output:** The final estimate θ_k as the MLE.

R Code Implementation:

Fisher Scoring Function

```

1 fisher_scoring_mle <- function(x, tol = 1e-5, max_iter = 1000) {
2   n <- length(x)
3   theta_k <- mean(x)
4   error <- Inf
5   iter <- 0
6   while (error > tol && iter < max_iter) {
7     sum_term <- sum(1 / (1 + exp(x - theta_k)))
8     theta_next <- theta_k + 3 - (6 / n) * sum_term
9
10    error <- abs(theta_next - theta_k)
11    theta_k <- theta_next
12    iter <- iter + 1
13  }
14  if (iter == max_iter) {
15    warning("Maximum iterations reached without convergence.")
16  }
17  return(list(mle = theta_k, iterations = iter))
18 }

```

- (f) Let $\theta = 0$. Consider the following R code to generate 100 data points for Logistic(0, 1) distribution and then calculating MLE using the above algorithm

Sampling and Estimating θ

```

1 sample_size <- 100
2 t <- 0
3 sample_0 <- rlogis(sample_size, t, 1)
4
5 fisher_scoring_mle(sample_0)

```

Thus, we obtain the following MLE for θ

$$\hat{\theta}_{ML} = 0.02623463$$

which is very close to the actual value of θ , i.e., 0

(g) Consider the following R code for simulating the results for a 1000 times

Simulating 1000 MLE Estimates ($n = 100$)

```
1 sample_size <- 100
2 t <- 0
3 num_sims <- 1000
4 estimates <- c(num_sims)
5 for (i in 1:num_sims) {
6   sample_0 <- rlogis(sample_size, t, 1)
7   estimates[i] = fisher_scoring_mle(sample_0)$mle
8 }
9 hist(estimates)
```

Now we have a list named *estimates* which has 1000 MLE estimates for different samples of the same distribution. The histogram for the same has been presented below

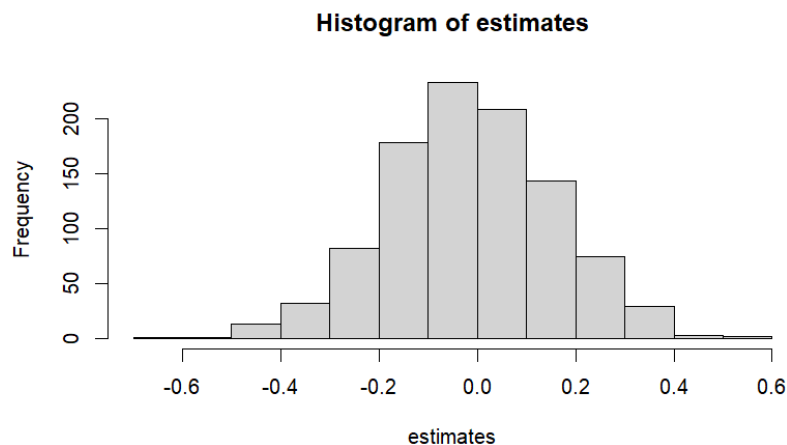


Figure 1: Histogram of 1000 MLE estimates for θ with sample size = 100.

The histogram is unimodal, roughly symmetric, and centered right around the true value $\theta = 0$, indicating that the MLE is (at least approximately) unbiased. Combined with its reasonably tight spread, this suggests the MLE is a good estimator of θ .

(h) We'll be repeating parts (f) and (g) for $\theta = 10$ now. Thus, we have the following

- For $n = 50$. Consider the following R Code and results

Simulating 1000 MLE Estimates ($n = 50$)

```
1 sample_size <- 50
2 t <- 10
3 estimates <- c(num_sims)
```

```

4 for (i in 1:num_sims) {
5   sample_0 <- rlogis(sample_size, t, 1)
6   estimates[i] = fisher_scoring_mle(sample_0)$mle
7 }
8 hist(estimates)

```

For a single simulation, we have the following ML estimate

$$\hat{\theta}_{ML} = 10.01711$$

For a 1000 simulation, we have the following histogram

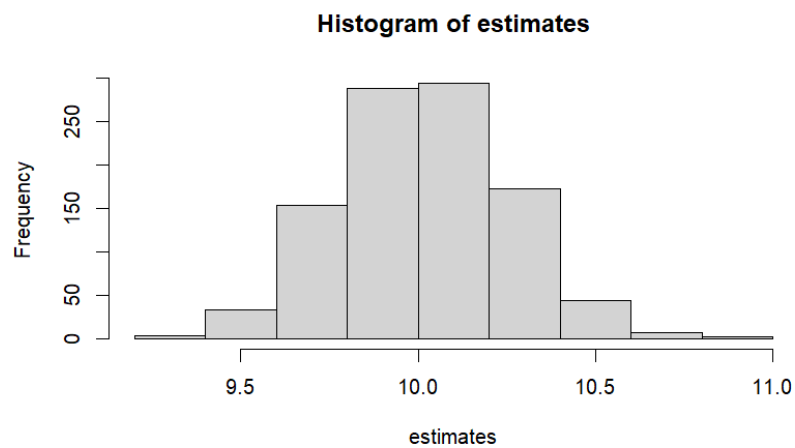


Figure 2: Histogram of 1000 MLE estimates for θ with sample size = 50.

- For $n = 200$. Consider the following R Code and results

Simulating 1000 MLE Estimates ($n = 200$)

```

1 sample_size <- 200
2 t <- 10
3 estimates <- c(num_sims)
4 for (i in 1:num_sims) {
5   sample_0 <- rlogis(sample_size, t, 1)
6   estimates[i] = fisher_scoring_mle(sample_0)$mle
7 }
8 hist(estimates)

```

For a single simulation, we have the following ML estimate

$$\hat{\theta}_{ML} = 10.0006$$

For a 1000 simulation, we have the following histogram

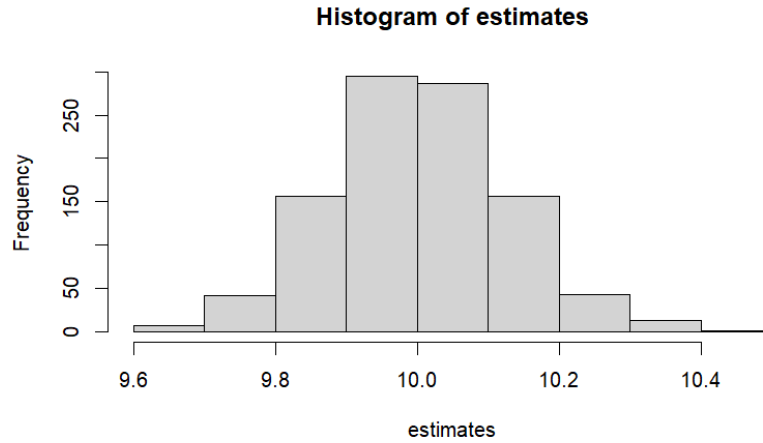


Figure 3: Histogram of 1000 MLE estimates for θ with sample size = 200.

- For $n = 500$. Consider the following R Code and results

Simulating 1000 MLE Estimates ($n = 500$)

```

1 sample_size <- 500
2 t <- 10
3 estimates <- c(num_sims)
4 for (i in 1:num_sims) {
5   sample_0 <- rlogis(sample_size, t, 1)
6   estimates[i] = fisher_scoring_mle(sample_0)$mle
7 }
8 hist(estimates)

```

For a single simulation, we have the following ML estimate

$$\hat{\theta}_{ML} = 9.997625$$

For a 1000 simulation, we have the following histogram

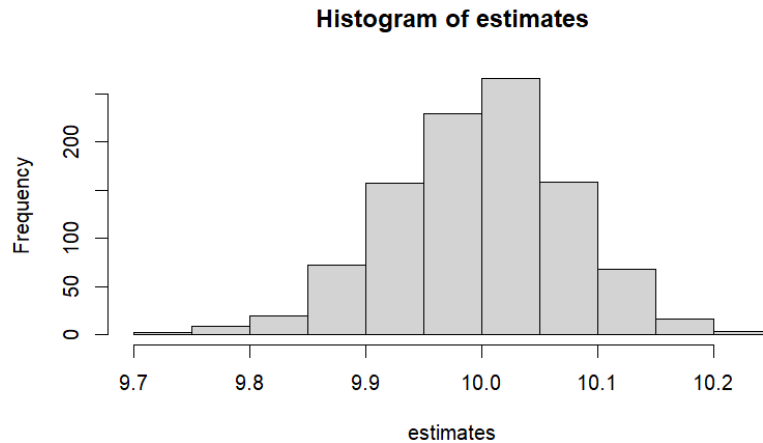


Figure 4: Histogram of 1000 MLE estimates for θ with sample size = 500.

Across $n = 50, 200, 500$, each histogram stays centered at the true value $\theta = 10$, but visibly narrows as n grows. This shrinking spread confirms that the MLE is consistent, its variance decreasing towards 0 as the sample size increases.